

- 1) Print the following pattern without using any array or data structures.

Input:

5

Output:

```
1  3  6 10 15
2  5  9 14
4  8 13
7 12
11
```

- 2) Print the following array sorting with only two loops.

Input:

34 6 2 4 12 8 9 1 23

Output:

34 1 23 2 12 4 9 6 8

Even positions Should be in descending order arranged and odd positions should be in the ascending order arranged.

- 3) Find the Longest Palindrome Substring in the given string.

Input:

ganadanam

Output:

anadana

Input:

lirilsoap

Output:

liril

Input:

amadam

Output:

madam

Input:

Shoe

Output:

There is no such palindrome in the given string.

- 4) Print the top 3 repeated elements in the given array.

Input:

3 15 6 34 3 6 15 3 3 6

Output:

3
6
15

- 5) Print the sum for the given integer with the sum of minimum number of binary-decimal numbers.

Input:

32

Output:

11 + 11 + 10

Count:

3

Explanation:

32 can be made with multiple combinations like

10 + 10 + 10 + 1 + 1

Which has count 5 which is not the minimum number of combinations.